

UTILIZING GPG AND VERACRYPT FOR ENCRYPTION OF FILES AND MESSAGES

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PROJECT OVERVIEW

This project demonstrates hands-on implementation of encryption technologies using GPG (GNU Privacy Guard) on Kali Linux for file-level asymmetric encryption and VeraCrypt for container-based symmetric encryption. The objective was to understand encryption mechanisms used to protect sensitive data at rest and during secure file handling.

GPG ENCRYPTION

```
(drey@DREY)-[~]
$ gpg --full-generate-key
gpg (GnuPG) 2.4.8; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: keybox '/home/drey/.gnupg/pubring.kbx' created
Please select what kind of key you want:
(1) RSA and RSA
(2) DSA and Elgamal
(3) DSA (sign only)
(4) RSA (sign only)
(9) ECC (sign and encrypt) *default*
(10) ECC (sign only)
(14) Existing key from card
Your selection? 9
Please select which elliptic curve you want:
(1) Curve 25519 *default*
(4) NIST P-384
(6) Brainpool P-256
Your selection? 1
Please specify how long the key should be valid.
 0 = key does not expire
<n> = key expires in n days
<n>w = key expires in n weeks
<n>m = key expires in n months
<n>y = key expires in n years
Key is valid for? (0) 3
Key expires at Sun 08 Feb 2026 06:30:57 AM EST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: david
Email address: davidapena101@gmail.com
Comment: this is a gpg encryption pratice
You selected this USER-ID:
"david (this is a gpg encryption pratice) <davidapena101@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/drey/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/drey/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/drey/.gnupg/openpgp-revocs.d/A34B20A589ABDD0F5515A97E5505D6E2AE498CFB.rev'
public and secret key created and signed.

pub   ed25519 2026-02-05 [SC] [expires: 2026-02-08]
      A34B20A589ABDD0F5515A97E5505D6E2AE498CFB
uid     david (this is a gpg encryption pratice) <davidapena101@gmail.com>
sub   cv25519 2026-02-05 [E] [expires: 2026-02-08]
```

On Kali linux terminal, A public key pair was generated using the command ***“gpg --full-generate-key”*** and the default sub-settings. In here the validity period of the key was selected, the email address and password was created

```

(drey@DREY)-[~]
$ pwd
/home/drey
(drey@DREY)-[~]
$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
(drey@DREY)-[~]
$ ls -la
total 144
drwx----- 16 drey drey 4096 Feb 5 07:05 -
drwxr-xr-x 3 root root 4096 Oct 7 06:53 --
-rw-r--r-- 1 drey drey 220 Oct 7 06:53 .bash_logout
-rw-r--r-- 1 drey drey 551 Oct 7 06:53 .bashrc
-rw-r--r-- 1 drey drey 3526 Oct 7 06:53 .bashrc.original
drwxrwxr-x 12 drey drey 4096 Jan 30 06:58 .cache
drwxr-xr-x 14 drey drey 4096 Feb 5 06:33 .config
drwxr-xr-x 2 drey drey 4096 Feb 5 06:51 Desktop
-rw-r--r-- 1 drey drey 35 Oct 7 06:57 .dmrc
drwxr-xr-x 2 drey drey 4096 Oct 7 06:57 Documents
drwxr-xr-x 2 drey drey 4096 Jan 30 05:50 Downloads
-rw-r--r-- 1 drey drey 11759 Oct 7 06:53 .face
lrwxrwxrwx 1 drey drey 5 Oct 7 06:53 .face.icon -> .face
drwx 4 drey drey 4096 Feb 5 06:37 .gnupg
-rw 1 drey drey 0 Oct 7 06:57 .ICEauthority
drwxr-xr-x 3 drey drey 4096 Oct 7 06:53 .java
drwxr-xr-x 5 drey drey 4096 Oct 7 06:57 .local
drwx 4 drey drey 4096 Jan 27 05:36 .mozilla
drwxr-xr-x 2 drey drey 4096 Oct 7 06:57 Music
drwxr-xr-x 2 drey drey 4096 Feb 5 06:40 Pictures
-rw-r--r-- 1 drey drey 807 Oct 7 06:53 .profile
drwxr-xr-x 2 drey drey 4096 Oct 7 06:57 Public
-rw-r--r-- 1 drey drey 0 Oct 7 07:13 .sudo_as_admin_successful
drwxr-xr-x 2 drey drey 4096 Oct 7 06:57 Templates
drwxr-xr-x 2 drey drey 4096 Oct 7 06:57 Videos
-rw 1 drey drey 49 Feb 5 06:22 .Xauthority
-rw 1 drey drey 10014 Feb 5 06:51 .xsession-errors
-rw-r--r-- 1 drey drey 4411 Feb 4 07:41 .xsession-errors.old
-rw-r--r-- 1 drey drey 336 Oct 7 06:53 .zprofile
-rw 1 drey drey 431 Feb 5 07:05 .zsh_history
-rw-r--r-- 1 drey drey 10855 Oct 7 06:53 .zshrc
(drey@DREY)-[~]
$ cd /home/drey/.gnupg/
(drey@DREY)-[~/gnupg]
$ ls
openpgp-revocs.d  private-keys-v1.d  pubring.kbx  pubring.kbx~  trustdb.gpg
(drey@DREY)-[~/gnupg]
$ cd openpgp-revocs.d
cd openpgp-revocs.d: command not found
(drey@DREY)-[~/gnupg]
$ cd openpgp-revocs.d

```

Using “*openpgp-revocs.d*” folder inside the “*gnupg*” folder, I have been able to successfully create new directory and also get the created public key.

```

(drey@DREY)-[~/gnupg]
$ cd openpgp-revocs.d
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ ls
A34B20A589ABDD0F5515A97E5505D6E2AE498CFB.rev
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ gpg --armor --export davidapena101@gmail.com > pubkey.asc
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ ls
A34B20A589ABDD0F5515A97E5505D6E2AE498CFB.rev  pubkey.asc
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ ls
A34B20A589ABDD0F5515A97E5505D6E2AE498CFB.rev  pubkey.asc
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ touch drey_encrypt.txt
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ ls
A34B20A589ABDD0F5515A97E5505D6E2AE498CFB.rev  drey_encrypt.txt  pubkey.asc
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ cat drey_encrypt.txt
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ echo 'I am an aspiring cybersecurity professional with a background in chemistry, currently learning ethical hacking and security fundamentals. This repository showcases my hands-on labs, tools, and learning projects as I build practical cybersecurity skills' >> drey_encrypt.txt
(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ cat drey_encrypt.txt
I am an aspiring cybersecurity professional with a background in chemistry, currently learning ethical hacking and security fundamentals. This repository showcases my hands-on labs, tools, and learning projects as I build practical cybersecurity skills

```

To export the public key to the email, ‘*gpg – armour –export*’ was used and command prompt ‘*touch*’ was used to create a file ‘*drey_encrypt.txt*’. Then ‘*echo*’ was used to make an input of the message to be encrypted into the file

```

(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ cat drey_encrypt.txt
I am an aspiring cybersecurity professional with a background in chemistry, currently learning ethical hacking and security fundamentals. This repository showcases my hands-on labs, tools, and learning projects as I build practical cybersecurity skills

(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$

(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ cat drey_encrypt.txt.gpg
^++++[<~@+ue
n~\~-m++++D*Y0+rI+
++%++'R++J+s+ v{+|+++?Gæ+u+Q+g+LD8++18+3)*N++BJCe9<ôHT+)*+u+v+g8++Q+BzJN+++ $ ++++Z++i+e:++++S+AQ4 6r++
g+1++++d+=6-H+o+qu}+kS+++_Qk{++!T+rY+++W+/z+H+m+_oL'+Q8+++~+uP+Y+++W++++3>+++R++++;c<+++++
L+kj++O++k+++«S++M++X++
J+âc!++S++++

(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ ls
A34B20A589ABDD0F5515A97E5505D6E2AE498CFB.rev drey_encrypt.txt drey_encrypt.txt.gpg pubkey.asc

(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ gpg --decrypt drey_encrypt.txt.gpg
gpg: encrypted with cv25519 key, ID B2DD19A9F05B3C7E, created 2026-02-05
"dauid (this is a gpg encryption pratice) <davidapena101@gmail.com>"

gpg: public key decryption failed: No passphrase given
gpg: decryption failed: No passphrase given

```

Using ‘*cat*’ you can view the message in the file before encryption, but after using ‘*gpg --encrypt --recipient davidapena100@gmail.com drey_encrypt.txt*’ to encrypt when you ‘*cat*’ the file again the message isn’t visible until you decrypt. While decrypting we got a failed feedback because the password was incorrect

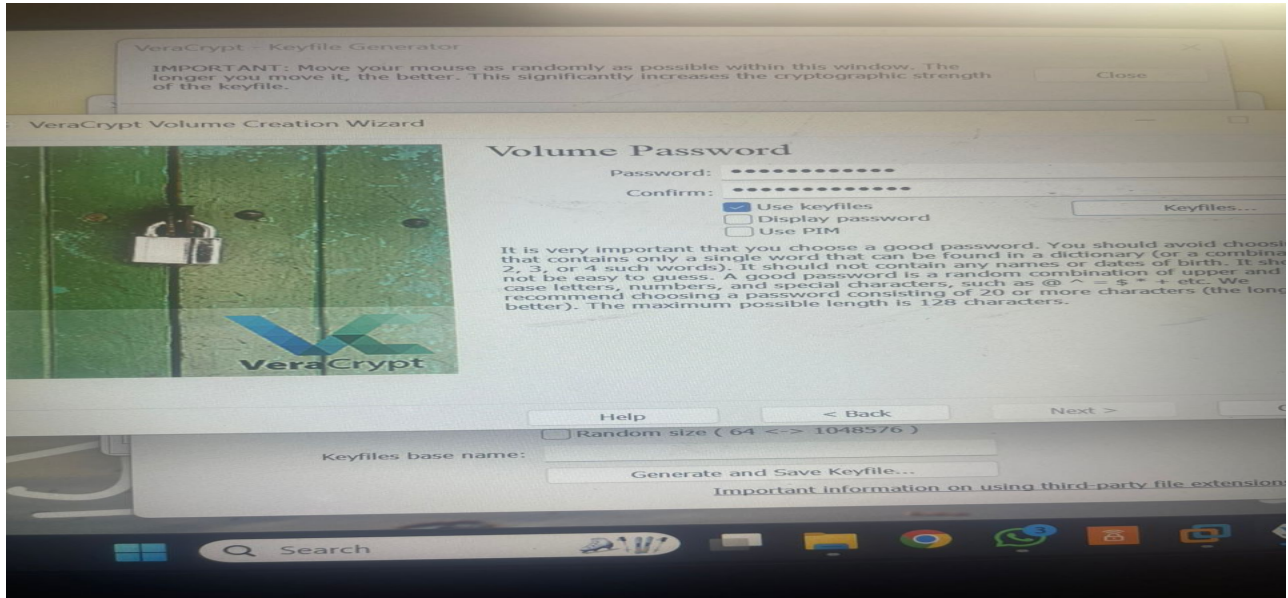
```

(drey@DREY)-[~/gnupg/openpgp-revocs.d]
$ gpg --decrypt drey_encrypt.txt.gpg
gpg: encrypted with cv25519 key, ID B2DD19A9F05B3C7E, created 2026-02-05
"dauid (this is a gpg encryption pratice) <davidapena101@gmail.com>"
I am an aspiring cybersecurity professional with a background in chemistry, currently learning ethical hacking and security fundamentals. This repository showcase
s my hands-on labs, tools, and learning projects as I build practical cybersecurity skills

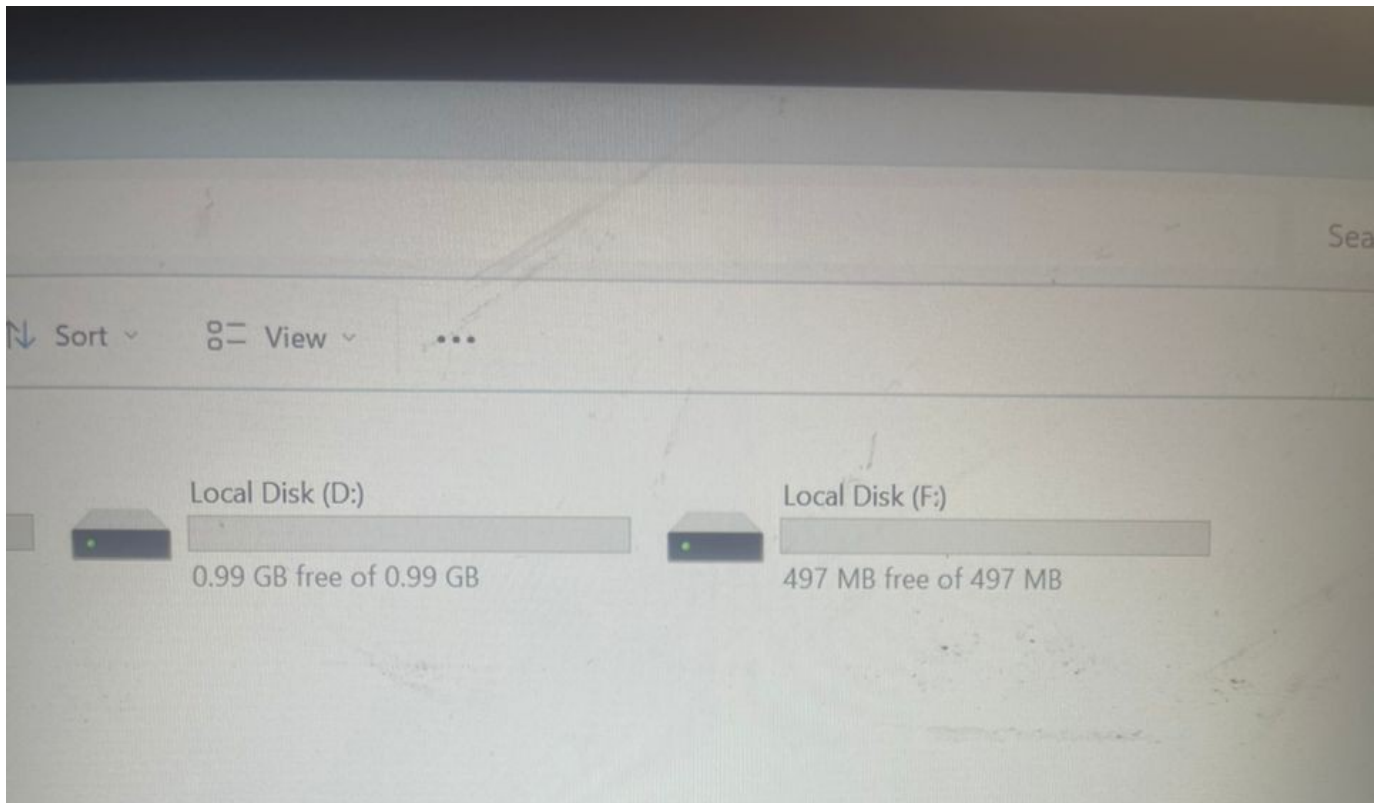
```

This shows a successful Decryption using ‘*gpg --decrypt drey_encrypt.txt.gpg*’.

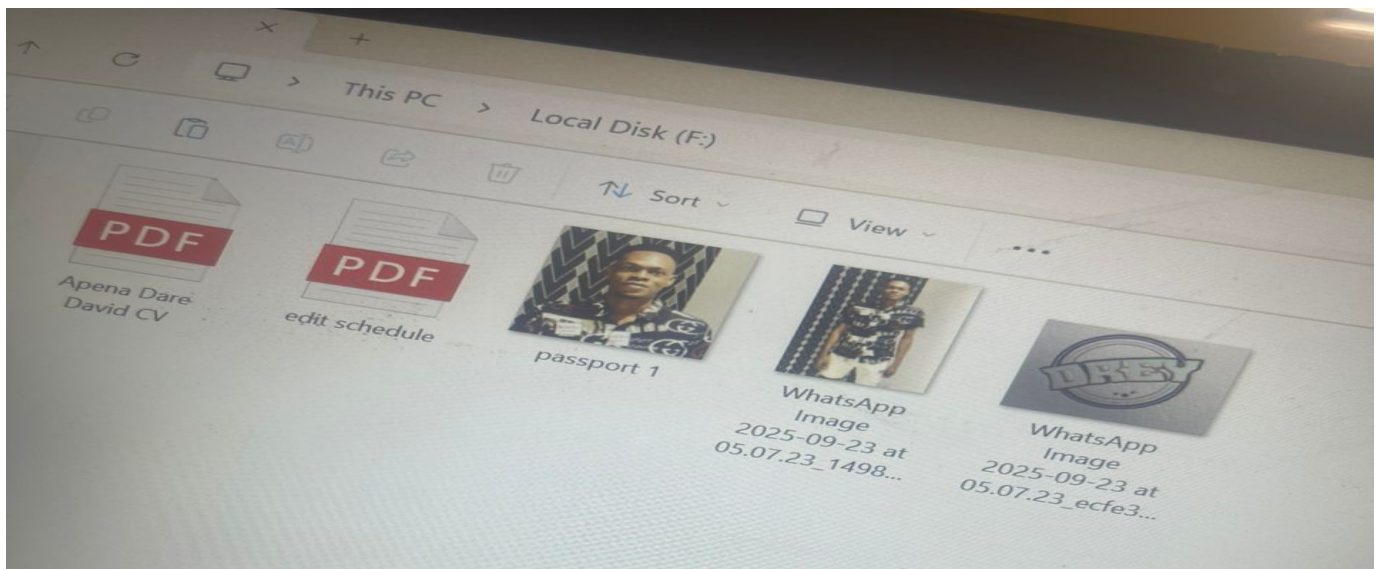
VERACRYPT



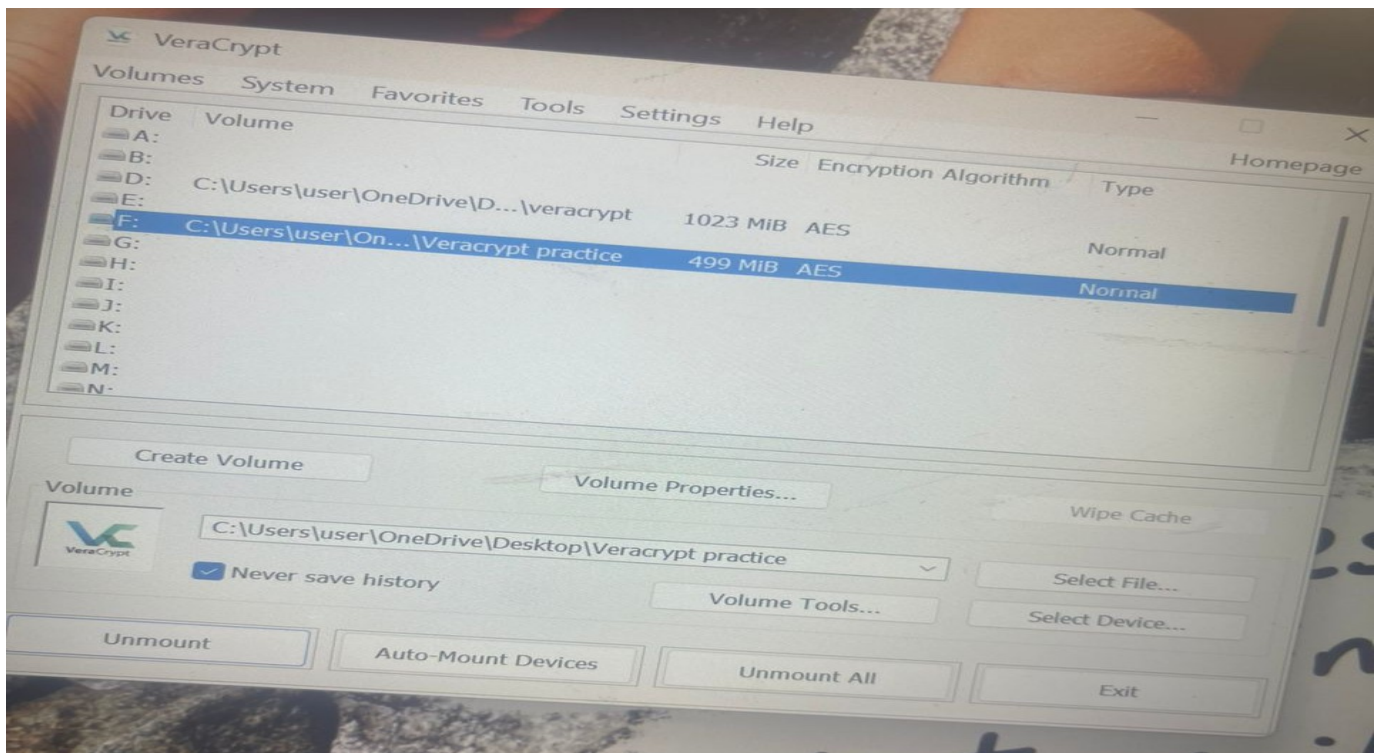
After the downloading and installation of VeraCrypt a password was created



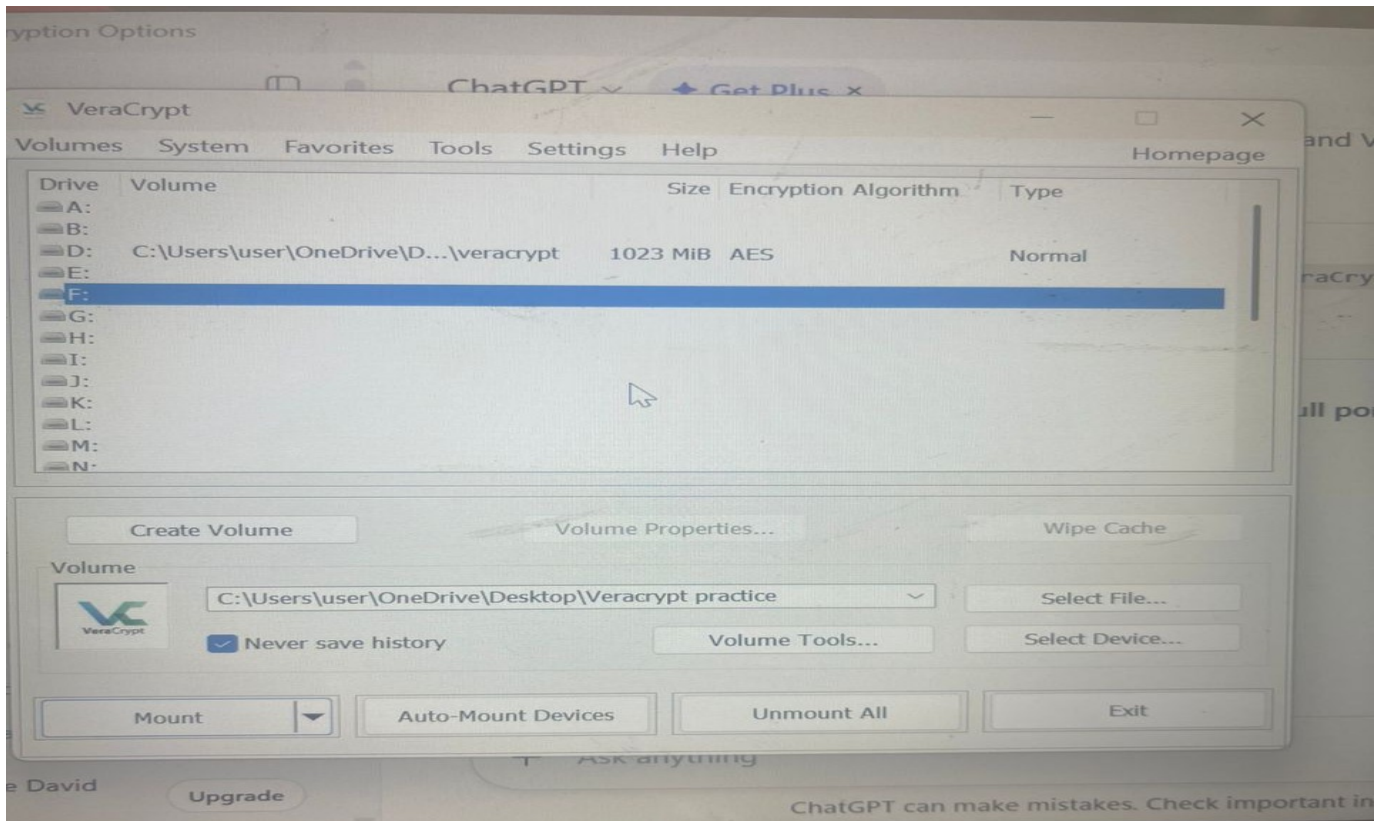
This is where the folder and container creation took place



Document and pictures to be encrypted were moved to the folder



The file was mounted, Mounting decrypts container in memory



After that it was unmounted, it removes access to view the file. To view again all you have to do is mount it.

TECHNICAL COMPARISON SUMMARY

GPG uses asymmetric encryption with public/private key pairs and is best suited for secure file exchange and digital signatures, while VeraCrypt uses symmetric encryption based on password-derived keys and is best suited for protecting stored data at rest within encrypted containers or volumes.

KEY SKILLS DEMONSTRATED

- Linux command-line proficiency
- Cryptography fundamentals (symmetric vs asymmetric)
- Secure key management
- Data-at-rest protection techniques
- Security documentation and lab reporting

ENVIRONMENT

- Kali Linux (VMware)
- GnuPG 2.4.x
- VeraCrypt
- Windows Host System

This project demonstrates practical implementation of both symmetric and asymmetric encryption technologies in real-world security workflows.

